

Summary of Key Findings

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This study investigated how the introduction of a variational quantum layer changes the learning dynamics of Physics-Informed Neural Networks (PINNs) for the Allen–Cahn and Heat equations. The objective was not to demonstrate universal superiority of QAPINNs over classical PINNs, but to understand when, why, and how quantum feature mappings modify the learning process.

The main findings are summarized as follows:

1. Quantum layers modify the representation space rather than simply improving accuracy.

The QAPINN architecture introduces a nonlinear quantum feature transformation between a classical encoder and decoder. This changes how the network represents the PDE solution. The experiments show that quantum layers can achieve competitive accuracy with significantly fewer trainable parameters, demonstrating their potential as compact feature mappings.

2. Parameter reduction is achieved at the cost of increased computational complexity.

For the Allen–Cahn equation, the classical PINN required approximately 1185 parameters, while the six-qubit one-layer QAPINN achieved the best observed performance with only 563 parameters. However, quantum simulation introduced a substantial increase in training time due to the computational cost of state-vector simulation and differentiable quantum circuit evaluation.

3. The optimal quantum circuit configuration is problem dependent.

Increasing the number of qubits generally improved the representation capacity for the nonlinear Allen–Cahn equation. The six-qubit one-layer architecture achieved the lowest observed loss among the tested configurations. However, increasing circuit depth beyond the required complexity degraded performance, indicating that additional expressivity can negatively affect optimization.

4. Circuit depth introduces a trade-off between expressivity and trainability.

Although deeper variational circuits provide a larger hypothesis space, they also introduce more optimization parameters and more complex loss landscapes. The experiments showed that shallow circuits are often preferable, and that increasing qubit number before increasing circuit depth is a more effective strategy for these PDE problems.

5. Data re-uploading and entanglement provide effective quantum feature extraction.

The employed quantum circuit uses repeated coordinate encoding through parameterized rotations and ring-based CNOT entanglement. This design allows the quantum layer to construct nonlinear feature transformations suitable for continuous PDE solutions while maintaining trainable gradient behavior for the investigated small-scale systems.

6. Quantum layers do not automatically improve generalization.

Although several QAPINN configurations improved interpolation accuracy, the extrapolation experiments demonstrated that higher expressivity does not necessarily translate into better prediction outside the training domain. Physical generalization remains strongly dependent on the architecture and training strategy.

7. A problem-specific design methodology is required.

The experiments indicate that QAPINNs should not use a universal circuit architecture. Instead, the number of qubits, circuit depth, encoding strategy, and measurement operators should be selected according to PDE characteristics, including nonlinearity, solution smoothness, and required frequency content.

Overall, the results demonstrate that the quantum layer acts as an alternative nonlinear feature representation mechanism for PINNs. Its main benefits are parameter efficiency and additional representational flexibility, while its main limitations are computational overhead and sensitivity to circuit design. The most effective QAPINN architectures are therefore expected to be shallow, problem-adapted quantum models rather than deeper circuits with maximum expressivity.